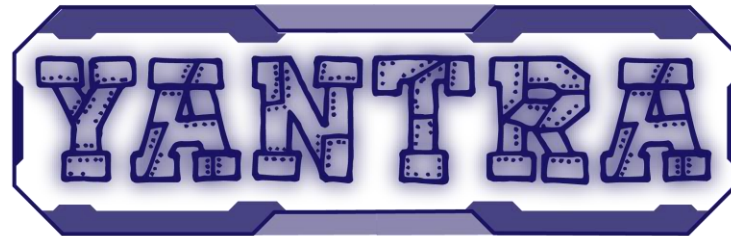


Indirect Rapid tooling

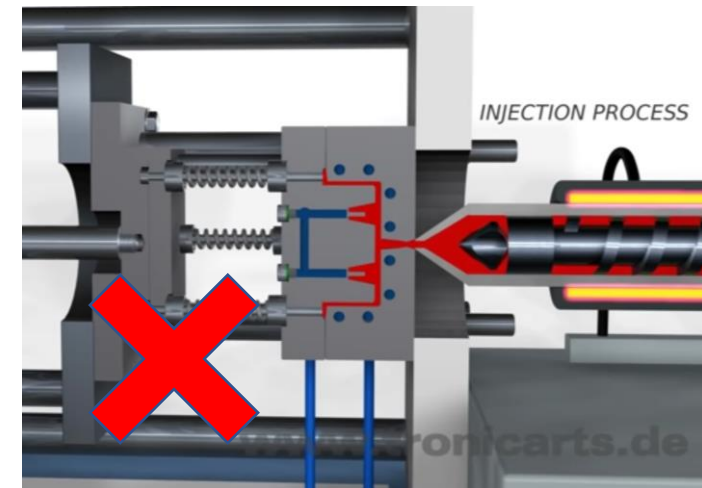


Silicone Rubber Tooling

- This is a sub-tooling technique - a rubber mould is prepared.
- This mould can be used **50 times** for preparing pattern or objective.
- An RP process is used to fabricate the pattern.
- The **pattern** is fix turned into a **holding box** a coated with a special release agent (wax based aerosol or petroleum jelly mixture) to prevent it from sticking to the silicon rubber mould.
- The **Silicon rubber** is then blended, vacuumed to remove air bubbles & **poured** into the box around the pattern until the pattern is completely encapsulated.

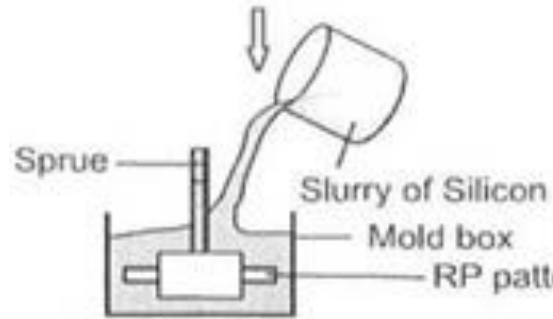


- After the rubber is fully cured (12-24 hours curing time), the box is removed & the **mould is cut into two halves** along a predetermined parting line.
- At this point the original pattern is cooled from the silicon rubber which can be placed back together & repeatedly filled with hot wax & plastic to **fabricate multiple patterns**.
- These tools are generally **not injected** due to **soft nature** of material so the final part material must be **poured** into the mould cavity per each cycle.

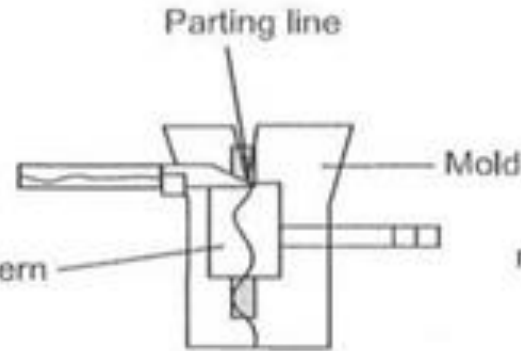


Silicone Rubber Tooling

Vacuum Chamber



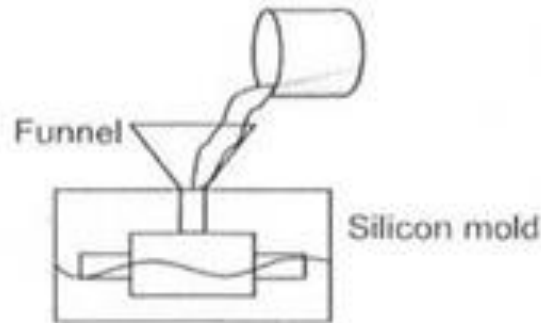
(a) Preparing silicon mold



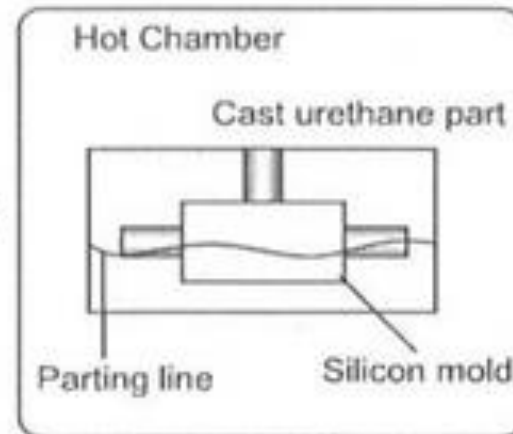
(b) RP Pattern



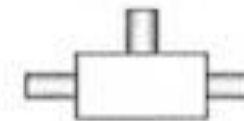
(c) Mixing of resin and catalyst



(d) Pouring resin mixture in to silicon mold



(e) Cast urethane part cured in a baking oven



(f) Final product

Side note, not to be studied



When a metal plastic needs to be embedded in plastic part, its called insert molding

Aluminium filled epoxy tooling

- Composite/Aluminium Filled Epoxy tooling is a rapid tooling technique that fills the void between current prototype processes.
- For **plastic parts**, the two most common tooling processes are

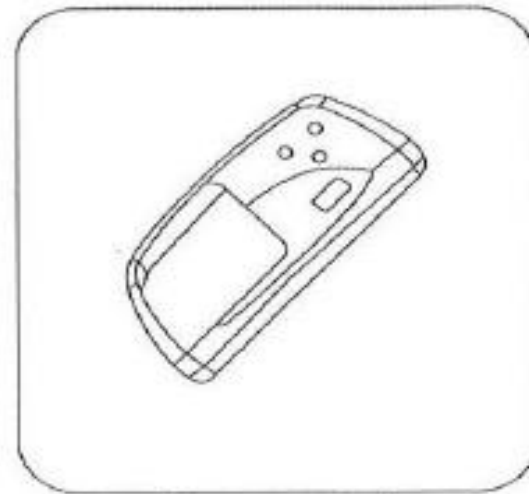
- 1. RTV moulding**

- 2. Soft tooling**

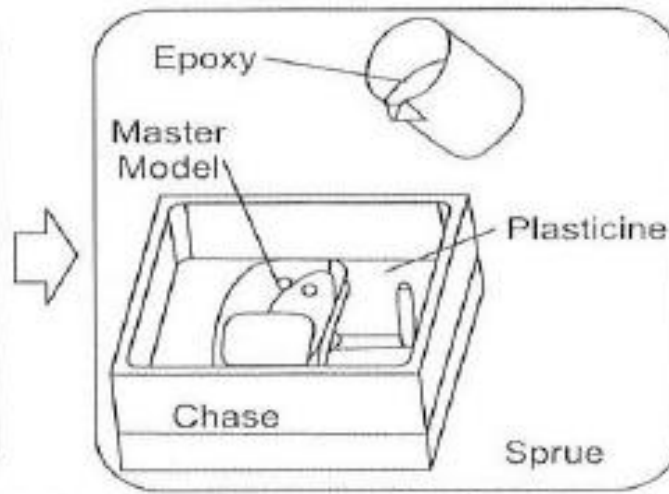
Composite tooling bridges this gap by offering **advantages** of both processes:

1. The **quicker time** and smaller expense associated with RTV moulding.
2. The capability of soft tooling to use production materials (thermoset plastics) for prototyping **injection moulded** parts.

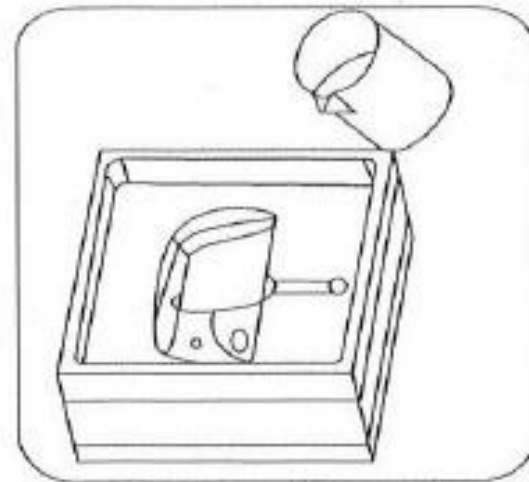
Aluminium filled epoxy tooling



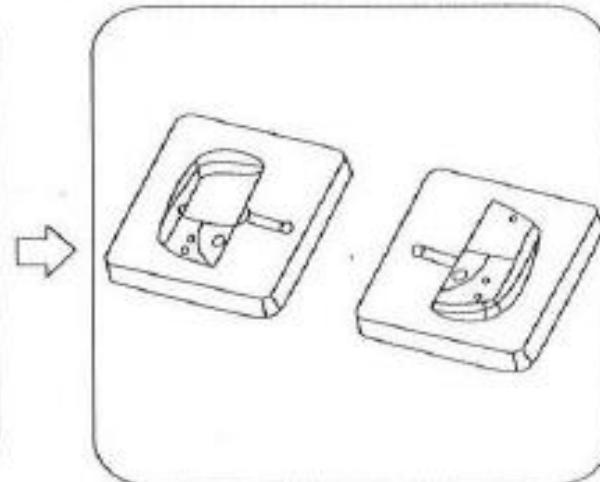
Master model



Embedding of the model in plasticine along its parting line and backfilling with epoxy



Inversion of the first half of the mould and backfilling

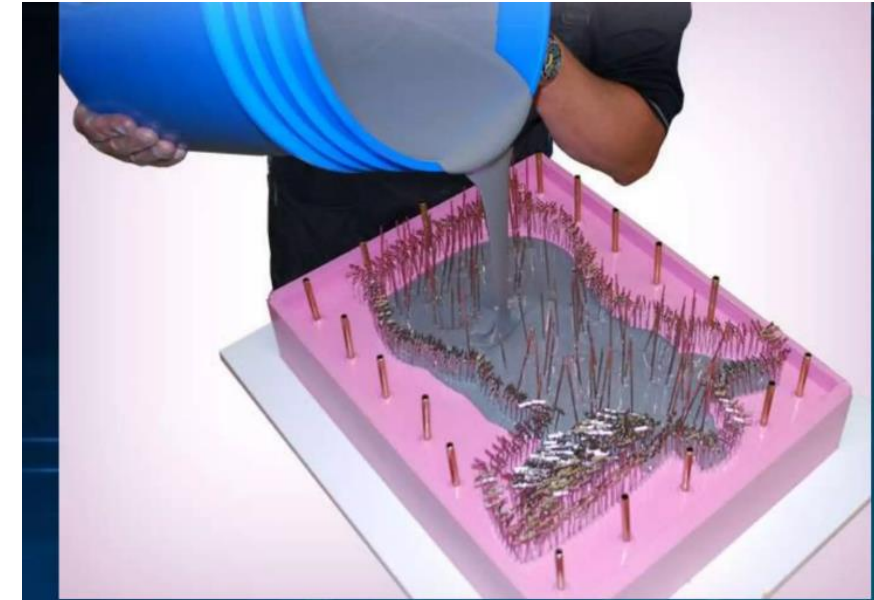


Epoxy injection moulding inserts

Aluminium filled epoxy tooling

RP generated **master pattern** is embedded in a wooden frame along with a parting line.

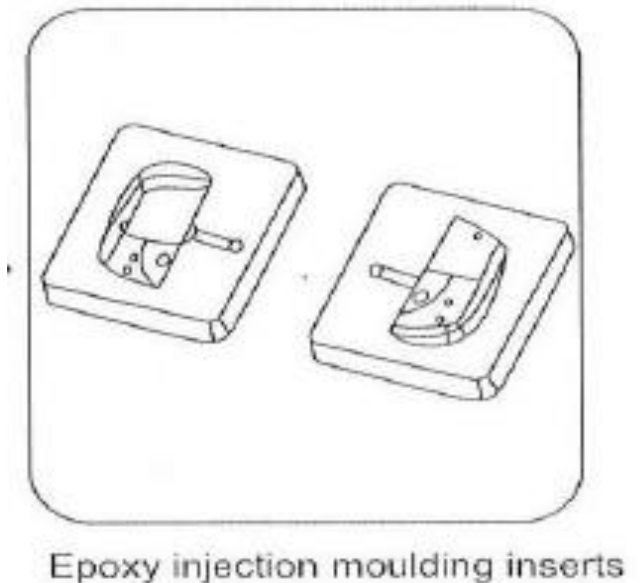
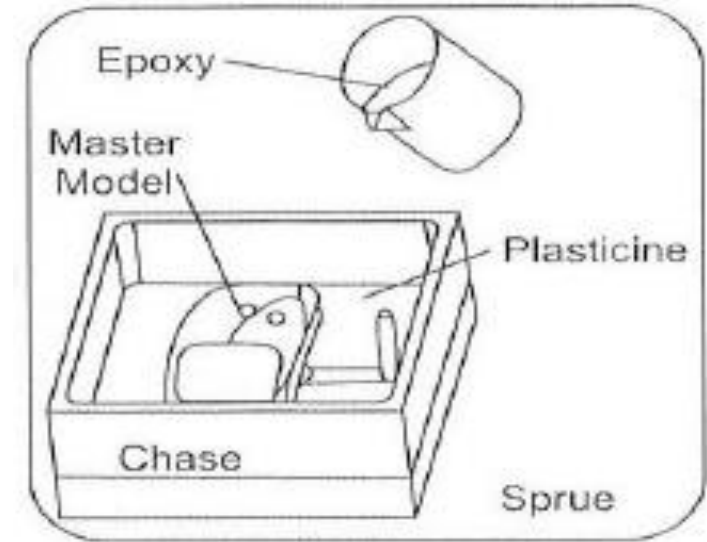
- **AL-filled epoxy** is poured around it to create the first half of the mould.
- It is usually **necessary to secure** the positioning of the **pattern** with specially fabricated wooden support & often specially machined metal inserts are placed in areas of the mould that might need strengthening.
- The parting line may also be a complex geometry rather than a simple plane, requiring additional fabrication step.



Aluminum filled epoxy tooling

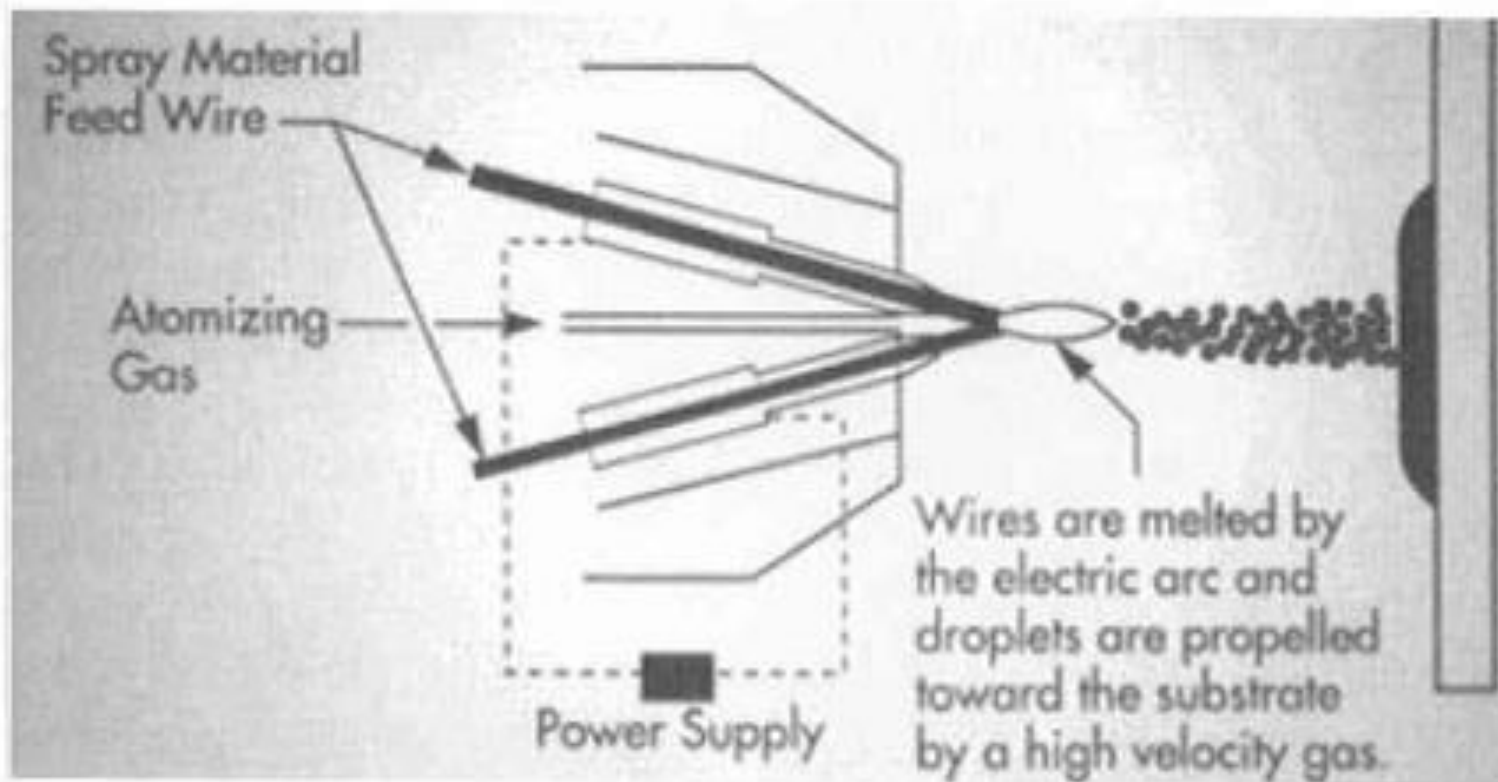


- **Metal inserts are placed** in areas where the epoxy is unlikely to withstand the pressures of the **injection-molding process**.
- After the mixture hardens, the entire **assembly is inverted** and the 2nd half of the mould is cast against the first.
- After the 2nd half of the mould is prepared the pattern is removed.
- **Aluminium filled epoxy** moulds are used in a mould frame.
- **Water cooling lines** are also being included during the fabrication process.
- This process works best for relatively **simple shapes**.



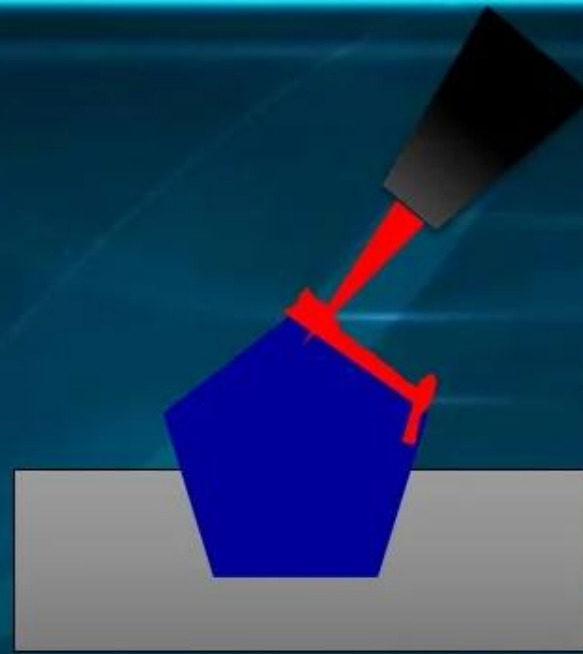
Spray Metal Tooling

- Thermal metal deposition technologies such as wire-arc spray and vacuum-plasma deposition are being developed to essentially coat low-temperature substrates with metallic materials.

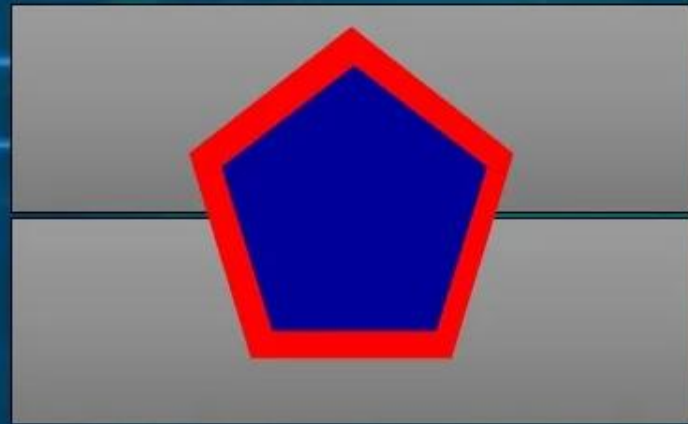




RP Model



Metal spraying

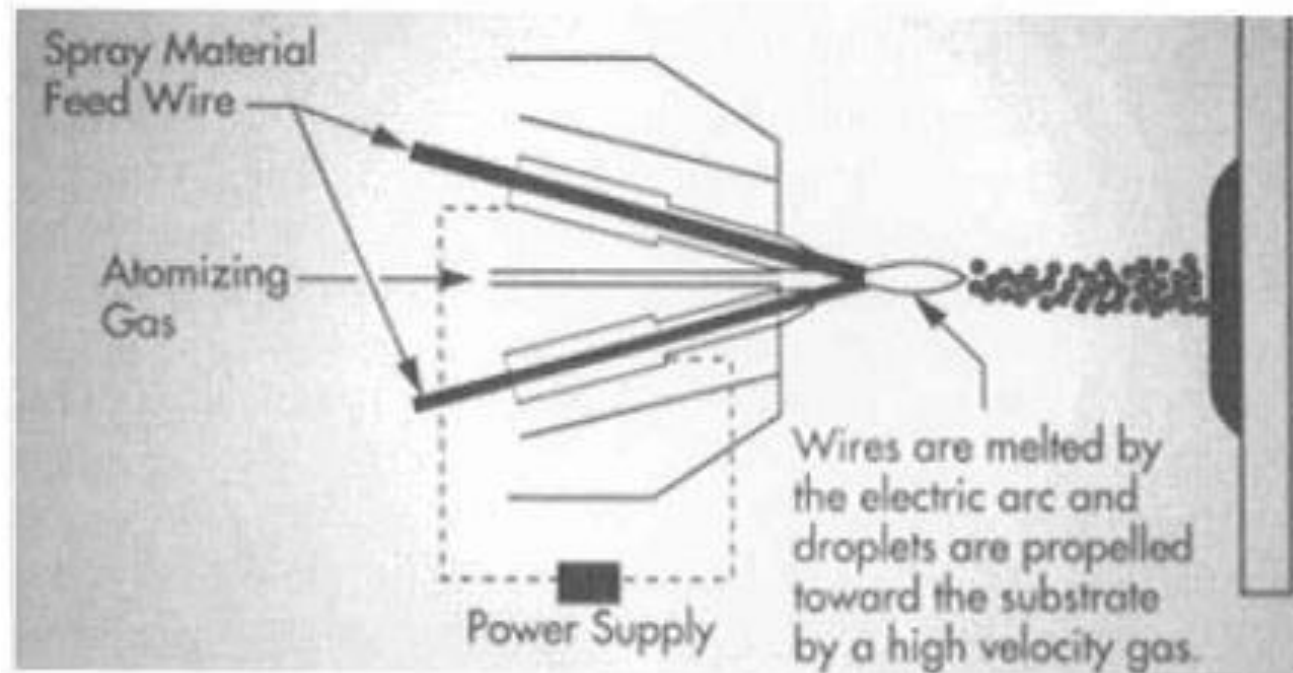


Finished Model

- The concept is to first deploy a high-temperature, high hardness shell material to an **RP positive**,
- and then backfill the remainder of the tool shell with **inexpensive low-strength, low temperature materials** and cooling channels (if necessary).
- This provides a **hard, durable face** that will endure the forces and temperatures of injection molding,
- and a soft backing that can be worked for optimal thermal conductivity and heat transfer from the mold.



- In wire-arc spray, the metal to be deposited comes in filament form.
- **Two filaments** are fed into the device, one is positively charged and the other negatively charged, until they meet and create an **electrical arc**.
- This arc melts the metal filaments, while simultaneously a high-velocity gas flows through the arc zone and propels the atomized metal particles onto the RP pattern.



- The spray pattern is either controlled **manually**, analogous to spray painting, or automatically by **robotic control**.
- Metal can be applied, in successive thin coats, to very low-temperature RP patterns without deformation of the geometry.
- Current wire-arc spray technologies are limited to **lower-temperature metals**, however, as well as to metals available in filament form.



Cast kirksite:

- It is advantageous for **complex geometries**.
- But it is generally **less accurate** and **more expensive** than Aluminium filled epoxy or spray metal tooling.
- Kirksite is a Zinc based alloy & the making process also starts with a RP generated pattern.
- Kirtsite tools have about the **same life** as spray metal or Aluminium filled epoxy tool.



3D KelTool:

- It is a method of creating moderate to high volume tool.
- This process was developed 25 years ago.
- It is presently owned & licensed by **3D system.**
- Any type of RP generated pattern can be used.
- It offers good accuracy & finish from a long lasting steel tool & many users have made more than a million parts from a Keltool mould.
- This process is based on metal sintering process

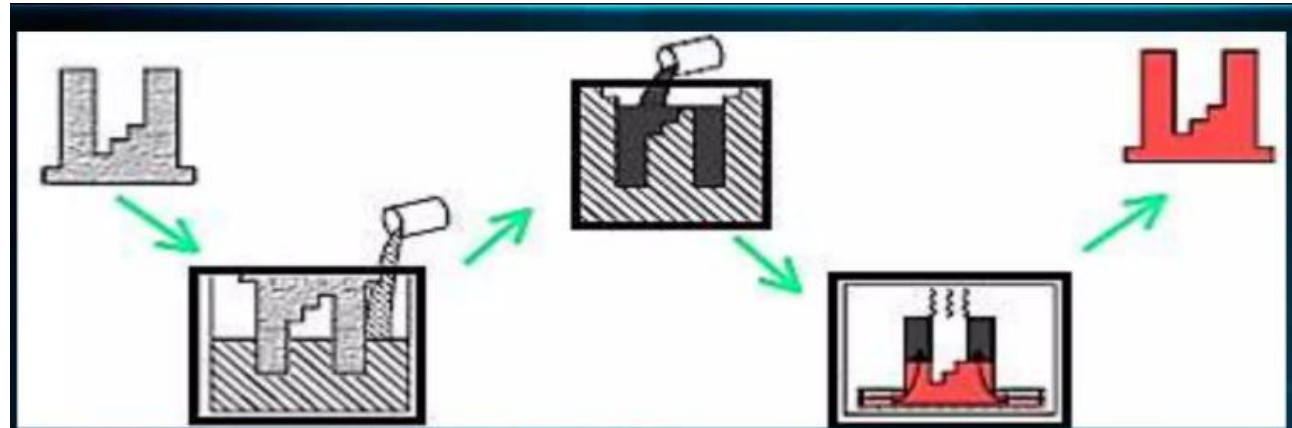


- The process starts with a RP pattern from which a rubber mould is created.
- The **rubber mould** is then used to cast (steel powder & polymer binder mixture) into the mould geometry which after hardening is in a green state.
- The **green mould** is fired and copper infiltrated resulting with a tool (70% steel & 30% copper).
- **Ejector pins, cooling lines** & other accessories may be added in final machining & polishing steps.



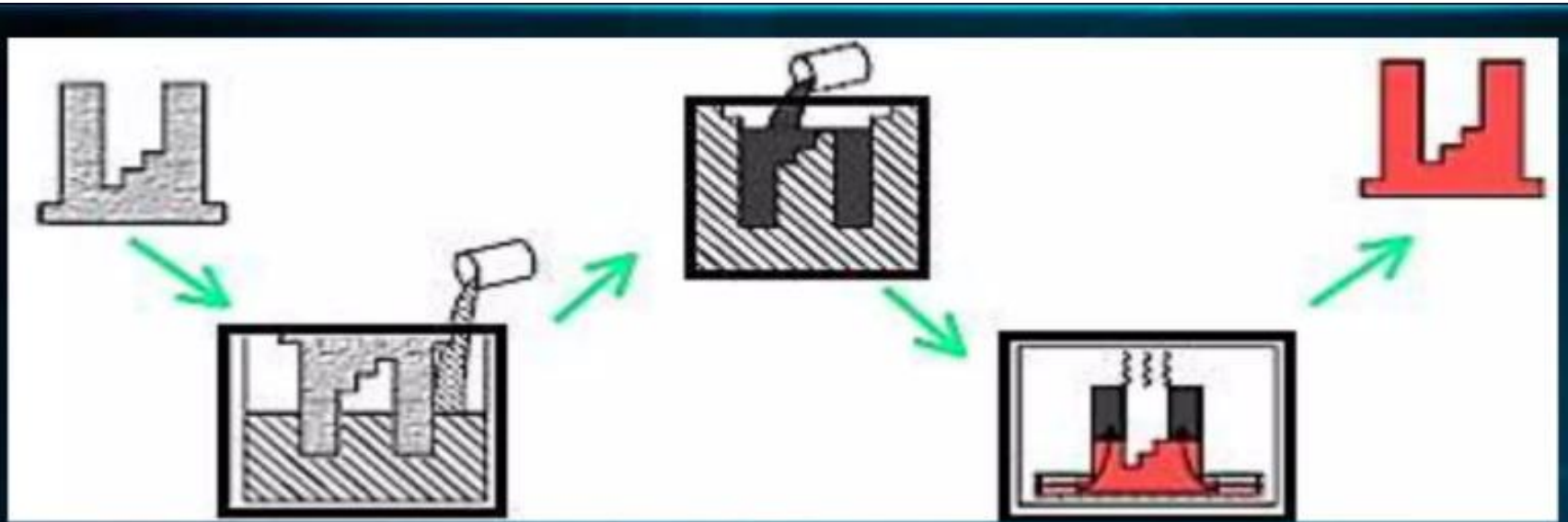
Ejector pins:
For ejecting
finished part

- In some cases little or no finishing is required.
- The inserts are strong enough to withstand typical injection moulding temperature and pressure condition & it is possible to use filled plastics & perform die casting with them.



Process chain of 3D Keltool:

- I. Master pattern
- II. Silicone casting
- III. Casting with a tool steel/Tungsten carbide/epoxy mixture
- IV. Burn-out of binder, sintering and infiltration with copper in an oven
- V. Tool insert ready for production



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3D Keltool

- The word "Keltool" refers to the proprietary powder metal sintering process, which involves penetrating a fused metal part with copper alloy.
- This process is based on metal sintering process.
- This process converts RP master patterns into production tool inserts with very good definition and surface finish.
- 3D Keltool inserts are built of Sterlite of A6 composite tool steel. The material properties allow the inserts using this process to withstand more than 10lakh mould cycles.
- The Keltool part (mold) is limited to 6 inches in all directions. Therefore, any parts produced would be limited to about 4 inches in all directions.

3D Keltool Process

- Fabricating the master patterns of core and cavity.
- Producing RTV silicon rubber mould from the pattern.
- Filling the silicon rubber mould with metal mixtures to produce green parts duplicating the masters.
- Firing the green parts in a furnace to remove the plastic binder and sintering the metal particles together.
- Infiltrating the sintered parts (70% dense inserts) with copper in the second furnace cycle to fill the 30% void space.
- Finishing the core and cavity.



SLA Master
Pattern of mold
geometry



RTV Transfer
Mold created



Fill Transfer Mold
with A6 Tool Steel
& Tungsten Carbide
Slurry



Sinter & infiltrate
with 30% copper